

Thematic analysis of 27 OCR posters

Raw text files were analyzed. Author and reference sections were ignored.

Category summary

Category	Count	Posters
Basic biology (Neurology)	2	Godoy-Pena et al. (2025) - CCC FA25.txt; Luera et al. (2025) - CCC FA25.txt
Basic biology (Metabolism)	7	Berbouti et al. (2025) - CCC FA25.txt; Camara et al. (2025) - LU SP25.txt; Dehuelbes et al. (2025) - LU SP5.txt; Lemus et al. (2025) - FA25 CCC.txt; Nii et al. (2025) - CCC FA25.txt; Pedireddi et al. (2025) - CCC SP25.txt; Tuttle et al. (2025) - CCC SP25.txt
Basic biology (Other)	6	Diaz et al. (2025) - CCC SP25.txt; Ford et al. (2023) - CCC SP23.txt; Henriquez et al. (2025) - COD WI24.txt; Holmes (2025) - CCC SP25.txt; Meraz et al. (2023) - CCC SP23.txt; Sakana et al. (2025) - CCC SP25.txt
Disease research (Neurology)	4	Avetisyan et al. (2023) - CCC SP23.txt; Cabral et al. (2025) - CCC FA25.txt; Gill & Alcazar (2025) - CCC FA25.txt; Haubelt & Alcazar et al. (2025) - CCC FA25.txt
Disease research (Metabolism)	5	Brown et al. (2025) - CCC SP25.txt; Ojala et al. (2025) - LU SP25.txt; Rodriguez (2025) - CCC SP25.txt; Trevino et al. (2022) - CCC SP22.txt; Zlaket et al. (2023) - CCC SP23.txt
Disease research (Other)	3	Logan et al. (2025) - CCC FA25.txt; Paderna et al. (2025) - CCC SP25.txt; Paramo-Ojeda et al. (2025) - CCC SP25.txt
Other	0	—

Poster-by-poster evidence

Avetisyan et al. (2023) - CCC SP23.txt

Category: Disease research (Neurology)

Note: Named autism-related genes and a gut-brain-axis framing make the disease motivation dominant.

Abstract: 'Genes such as dve, KCNQ, Ssadh, Hem, ebi, and glo ... have been identified as autism related.'

Introduction: 'The Gut-microbiome-brain axis (GUMBA)... has linked disorders such as Autism (ASD) to genes expressed within the midgut.'

Introduction: 'When communication between the midgut and the brain is disrupted, there is a higher probability of autism.'

Discussion: the poster links the results to 'genes, expressed in both the central nervous system and midgut'.

Berbouti et al. (2025) - CCC FA25.txt

Category: Basic biology (Metabolism)

Note: A digestion/energy-metabolism poster about trehalose breakdown and region-specific expression.

Title: 'Trehalase Expression in the Drosophila Midgut'.

Abstract: 'Trehalose is a critical sugar... in which its breakdown by trehalase provides glucose for energy metabolism.'

Introduction: 'Trehalose is a major sugar in insects. TREH breaks trehalose into glucose in the Drosophila midgut.'

Discussion: 'local metabolic or metal handling functions' and 'glucose transporter ... enriched in the far posterior digestive regions'.

Brown et al. (2025) - CCC SP25.txt

Category: Disease research (Metabolism)

Note: The poster is driven by acid reflux, a digestive disorder, even though it also uses gut-brain signaling language.

Title: 'Analysis Of Gene Expression Of Acids Reflux Related Genes Across The Midgut'.

Introduction: 'Acid reflux is a condition shown in millions of people in the world'.

bullet Introduction: 'To better comprehend the molecular mechanisms behind this condition'.

bullet Discussion: 'effects of acid reflux are influenced by genetic factors' and the work focuses on 'human-relevant genes associated with this condition'.

Cabral et al. (2025) - CCC FA25.txt

Category: Disease research (Neurology)

Note: MECP2, Rett syndrome, and neurodevelopmental dysfunction make this clearly disease-driven.

bullet Abstract: 'The methyl-CpG binding protein 2 (MECP2) gene is essential for neuronal nervous system.'

bullet Abstract: 'Mutations in MECP2 are linked to Rett syndrome and related neurodevelopmental disorders'.

bullet Introduction: 'MECP2 dysfunction is known to cause Rett syndrome and contribute to a range of neurological impairments'.

bullet Discussion: 'Because MECP2 is expressed across the brain, its loss disrupts activity'.

Camara et al. (2025) - LU SP25.txt

Category: Basic biology (Metabolism)

Note: The central question is how amylase genes support digestion across gut regions; disease references are secondary.

bullet Title: 'Functional Difference Between Amy-D, Amy-P and Amyrel in Drosophila Midgut'.

bullet Introduction: 'The Drosophila midgut has distinct regions and pH zones that may affect enzyme activity.'

bullet Introduction: the genes 'are orthologs to human amylases' and are tied only secondarily to 'diseases like diabetes and Alzheimer's'.

bullet Discussion: 'specialization of enzyme function along the midgut' and 'overlapping roles in digestion'.

Dehuelbes et al. (2025) - LU SP5.txt

Category: Basic biology (Metabolism)

Note: Energy homeostasis, fatty-acid metabolism, and feeding behavior dominate; no named disease is the main driver.

bullet Title: 'Regional Expression of Key Metabolic genes in Drosophila Midgut'.

bullet Introduction: 'Acetyl-CoA links carbohydrate, fat, & protein metabolism'.

bullet Introduction: 'CRAT is a mitochondrial enzyme involved in fatty acid metabolism' and 'may influence feeding behavior'.

bullet Discussion: 'Explore CRAT as a potential target for treating metabolic disorders in humans' is framed as a downstream implication rather than the core motivation.

Díaz et al. (2025) - CCC SP25.txt

Category: Basic biology (Other)

Note: This is about immune-defense biology and phagosome acidification, not a named disease model.

bullet Title: 'V-ATPase Gene Expression Reveals Cellular Defense Zones'.

bullet Abstract: 'To explore how different regions ... contribute to immune defense'.

bullet Introduction: 'Phagosomes are part of the cell's internal defense and cleanup system'.

bullet Discussion: the Cu region 'contributes to immune defense by nurturing an acidic environment'.

Ford et al. (2023) - CCC SP23.txt

Category: Basic biology (Other)

Note: An environmental-response / xenobiotic-metabolism poster rather than a disease poster.

bullet Title: 'Cytochrome P450 genes providing insecticide resistance'.

bullet Abstract: 'Some Cytochrome P450 genes play an important role in the development of insecticide resistance'.

bullet Introduction: 'the Drosophila melanogaster midgut is used as a model system for studying human gut function'.

bullet Discussion: 'genes involved in insecticide resistance ... are most highly expressed'.

Gill & Alcazar (2025) - CCC FA25.txt

Category: Disease research (Neurology)

Note: The explicit disease framing is Parkinson's and broader neuropsychiatric disorders.

Title: 'Differential Expression of Serotonin Receptor Genes Htr1a and Htr2a in the Prefrontal Cortex and Striatum'.

Abstract: 'related to the neuropsychiatric disorder Parkinson's Disease'.

Introduction: 'Many neuropsychiatric disorders including Parkinson's Disease, Schizophrenia, and Post-traumatic Stress Disorder'.

Discussion: 'stronger serotonergic signaling in the PFC circuits which are involved in cognition and emotional regulation'.

Godoy-Pena et al. (2025) - CCC FA25.txt

Category: Basic biology (Neurology)

Note: Sleep/circadian behavior and gut-brain signaling are the main focus, not a named disorder.

Title: 'Regulation of Sleep Quality: Analysis of Gene Expression'.

Abstract: 'determine what genes dictate the kind of sleep drosophila flies get'.

Abstract: 'genes known to influence the drosophila fly's circadian rhythm'.

Introduction: 'Sleep and diet play crucial roles in human health' and the study is aimed at sleep-quality genetics.

Haubelt & Alcazar et al. (2025) - CCC FA25.txt

Category: Disease research (Neurology)

Note: Parkinson's disease is named directly and drives the whole analysis of the dopaminergic synapse.

Title: 'Exploring DRD2: The Association of PD Risk with Dopamine Receptors'.

Abstract: 'related to the neuropsychiatric disorder Parkinson's Disease'.

Introduction: 'Many neuropsychiatric disorders including Parkinson's Disease... are seen to have direct correlation to dopamine receptor D2'.

Discussion: the poster links striatal dopamine pathways to Parkinson's symptoms such as tremor and rigidity.

Henriquez et al. (2025) - COD WI24.txt

Category: Basic biology (Other)

Note: A core physiology / immune-pathway expression study; disease relevance is only secondary.

Title: '6 Immune Pathways Exhibit Similar Patterns of Gene Expressions in Drosophila melanogaster Midgut'.

Abstract: 'Measurement of gene expression in six immune pathways... suggests similar levels of immune pathway protein expression'.

Introduction: 'The specific functions that each region performs have not been thoroughly characterized'.

Introduction: disease references such as Crohn's disease are presented as possible broader relevance, not the main motivation.

Holmes (2025) - CCC SP25.txt

Category: Basic biology (Other)

Note: A melanogenesis/pigmentation project that only later connects to Parkinson's and neurodegeneration.

Title: 'Melanogenesis within the Drosophila Midgut: Neuromelanin and Dopamine'.

Abstract: 'focused in genes connected to melanogenesis (synthesis of melanin) and its antioxidant uses'.

Abstract: the yellow gene is discussed for pigmentation and mating behavior, not a disease model.

Discussion: Parkinson's appears as a downstream implication of dysregulated neuromelanin rather than the main research driver.

Lemus et al. (2025) - FA25 CCC.txt

Category: Basic biology (Metabolism)

Note: This is a carbohydrate digestion / absorption study with human digestive relevance.

Title: 'Macromolecule Breakdown and Absorption'.

Abstract: 'Understanding differential gene expression... can uncover processes relevant to human digestion as well as metabolic disorders'.

Introduction: the anterior region is 'tasked with macromolecule breakdown' and the posterior region 'controls molecule absorption'.

Discussion: the work examines 'starch and sucrose' digestion and regional nutrient absorption.

Logan et al. (2025) - CCC FA25.txt

Category: Disease research (Other)

Note: The project is explicitly about cardiovascular disease and ACE-related human pathology, not neurology or metabolism.

Abstract: 'cardiovascular-related pathways and potential therapeutic directions'.

Introduction: 'Cardiovascular disorders are a prominent and widespread disease'.

Introduction: the human ACE gene is 'a critical regulator of blood pressure and fluid balance'.

Discussion: the poster frames the work as insight into 'human cardiovascular disease'.

Luera et al. (2025) - CCC FA25.txt

Category: Basic biology (Neurology)

Note: This is fundamentally a neuropeptide / feeding-behavior poster about the gut-brain axis.

Title: 'Region-Specific NPF/NPFR Signaling in the Drosophila Midgut'.

Abstract: 'explore their role in gut-mediated feeding behavior'.

Introduction: 'Neuropeptide F (NPF) is a conserved neuropeptide that regulates feeding, stress, and sleep behaviors'.

Discussion: 'local paracrine signaling' and 'feeding regulation' dominate the motivation.

Meraz et al. (2023) - CCC SP23.txt

Category: Basic biology (Other)

Note: This is a gene-function / tissue-patterning poster built around tyrosine kinase expression and cell growth.

Title: 'Tyrosine Kinase Expression in the Midgut Suggests a Role in the Copper Region'.

Abstract: 'Tyrosine kinases are enzymes related to cell growth and development'.

Introduction: the midgut expression pattern is used to infer gene function across subregions.

Discussion: the poster concludes that the genes 'have high specificity to the Cu region of the midgut'.

Nii et al. (2025) - CCC FA25.txt

Category: Basic biology (Metabolism)

Note: The center of gravity is iron-dependent regulation of fatty-acid synthesis and energy storage.

Title: 'Iron-Induced Modulation of Acetyl-CoA Carboxylase (ACC)... Impacts on Lipid Metabolism and Energy Storage'.

Abstract: 'ACC is the rate limiting enzyme that converts acetyl-CoA to malonyl-CoA, committing cells to fatty acid synthesis'.

Introduction: 'Iron is an essential micronutrient... [that] can disrupt oxidative balance and alter key metabolic pathways'.

Discussion: 'ACC expression is strongly enriched in the Fe region... associated with iron absorption and metal handling'.

Otala et al. (2025) - LU SP25.txt

Category: Disease research (Metabolism)

Note: The title and introduction explicitly make this a model for fatty-acid metabolism disorders.

Title: 'Fly Gut is a good model for fatty acid metabolism disorders'.

Introduction: 'Related disorders and symptoms include: MCADD and Carnitine transporter deficiency, hypoglycemia, liver dysfunction, and muscle weakness'.

Introduction: 'ACOX3 breaks down very long-chain fatty acids... dysfunction may lead to fat accumulation, liver disease, and inflammation'.

Discussion: links Acox3 to 'nutrient absorption and waste management' in the context of metabolic disorders.

Paderna et al. (2025) - CCC SP25.txt

Category: Disease research (Other)

Note: The poster is driven by albinism / melanogenesis, which is a named disorder but not a neurology or metabolism category.

bullet Title: 'Albinism and Melanogenesis Pathway'.

bullet Abstract: 'In humans, the gene SLC45A2 is associated with oculocutaneous albinism type 4'.

bullet Introduction: 'OCA4 can result from mutations in the gene SLC45A2'.

bullet Discussion: the poster keeps returning to 'melanin synthesis' and albinism-linked human genetics.

Paramo-Ojeda et al. (2025) - CCC SP25.txt

Category: Disease research (Other)

Note: This is a classic cancer-motivation poster centered on KRAS-driven colon cancer.

bullet Title: 'Ras85D Expression in Drosophila Midgut Highlights a Model for KRAS-Driven Colon Cancer'.

bullet Abstract: 'KRAS mutations drive many human cancers by promoting uncontrolled cell growth'.

bullet Introduction: 'Cancer is a complex disease driven by mutations in genes that regulate cell growth, survival, and DNA repair'.

bullet Discussion: the work frames Ras85D as a model for 'tumor initiation'.

Pedireddi et al. (2025) - CCC SP25.txt

Category: Basic biology (Metabolism)

Note: This is a carbohydrate-digestion poster, not a disease poster.

bullet Title: 'Beyond the Anterior: Amylase Gene Expression Suggests Widespread Polysaccharide Digestion'.

bullet Abstract: 'interested in genes involved in sugar metabolism'.

bullet Introduction: the project studies 'how Drosophila melanogaster digested polysaccharides'.

bullet Discussion: 'breakdown of polysaccharides' and 'nutrient absorption' are the central themes.

Rodriguez (2025) - CCC SP25.txt

Category: Disease research (Metabolism)

Note: Galactosemia is a named metabolic disorder and is the direct motivation for the work.

bullet Title: 'Differential Gene Expression of Galm1 in the Drosophila Midgut'.

bullet Abstract: 'Mutations in... GALM, are associated with the disease galactosemia'.

bullet Introduction: 'Galactosemia is a rare metabolic disorder'.

bullet Discussion: 'areas such as the p2_4 region in Drosophila and the kidneys in humans, are at a greater risk for damaging mutations'.

Sakana et al. (2025) - CCC SP25.txt

Category: Basic biology (Other)

Note: Developmental patterning and transcription-factor control are the core themes.

bullet Title: 'Differential Expression of the TALE Homeobox Genes in Drosophila Midgut patterning'.

bullet Abstract: 'TALE group of homeobox genes regulate cell and tissue-specific gene expression'.

bullet Introduction: 'homothorax ... known for its role in tissue patterning, stem cell maintenance, and organ development'.

bullet Discussion: 'Genes coding for homeodomain transcription factors play a key role in gene expression control'.

Trevino et al. (2022) - CCC SP22.txt

Category: Disease research (Metabolism)

Note: Zellweger Spectrum Disorder is a named metabolic/peroxisomal disorder and remains the main motivation.

bullet Title: 'Drosophila Melanogaster a Good Model System of Zellweger Spectrum Disorder'.

bullet Abstract: 'Zellweger Spectrum Disorder (ZSD) is a genetic disorder'.

bullet Introduction: 'ZSD is a rare inherited disorder characterized by the absence/reduction of functional peroxisomes'.

bullet Introduction: 'essential for beta-oxidation of very-long-chain fatty acids' and 'metabolic abnormalities'.

Tuttle et al. (2025) - CCC SP25.txt

Category: Basic biology (Metabolism)

Note: A metabolism / aging / pigmentation pathway study, not a disease model.

bullet Title: 'Gene Expression Patterns in the Tryptophan-Kynurenine Pathway'.

bullet Abstract: 'genes regulating Tryptophan and its metabolism pathways'.

bullet Introduction: 'tryptophan is metabolized through the kynurenine pathway'.

bullet Discussion: the project emphasizes 'regional specialization in TRY processing' and links to aging and energy metabolism.

Zlaket et al. (2023) - CCC SP23.txt

Category: Disease research (Metabolism)

Note: A peroxisomal / lipid-metabolism disease framing anchored by Zellweger syndrome.

bullet Title: 'ScpX: A Peroxisomal Gene and its Distinctive Paralogs'.

bullet Introduction: 'Errors within this gene can cause Zellweger's Syndrome'.

bullet Introduction: the gene functions in 'oxidation of fatty acid chains within peroxisomes' and cholesterol transport.

bullet Discussion: the research links ScpX/paralogs to 'peroxisomal function' and lipid/sterol pathways.

No poster was assigned to the catch-all Other category; all 27 fit one of the six substantive categories above.